Nelson Pairing for Closure Dynamics: Schrödinger-Type Evolution from Forward and Backward Closure Processes

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April 2026

Status: Preprint (not peer-reviewed)

Abstract

Paper XVII proved that generators of the form $L_{\mathrm{BK}} + iM$ — the backward Kolmogorov generator plus any imaginary potential — are necessarily dissipative and cannot produce unitary evolution. It identified the Nelson-type forward/backward pairing as the most promising route past this obstruction. The present paper carries out that construction within the closure framework.

Starting from the forward closure Langevin process and its time-reversal, the current and osmotic velocities are defined and shown to produce an exact continuity equation. Because the closure drift is a gradient, the current velocity is automatically irrotational, and a phase function S is determined without additional assumptions. The closure Madelung pair is derived: the continuity equation plus a modified Hamilton–Jacobi equation with an explicit quantum-potential-like term.

Recombining ρ and S into a Nelson wavefunction $\Psi_N = \sqrt{\rho}\, e^{iS/\sigma^2}$, the paper derives the *closure-Nelson wave equation*: a nonlinear Schrödinger equation in which the potential depends on the state through $\Delta|\Psi_N|/|\Psi_N|$. The central result is two-tiered: at the Paper XIV equilibrium, the nonlinear potential evaluates exactly so that the wave equation reduces to a Schrödinger equation with Witten Hamiltonian; for perturbations around equilibrium, this linear Schrödinger dynamics persists to leading order.

This constitutes the first paired-process derivation of Schrödinger-type dynamics in the equilibrium and near-equilibrium regime within the closure framework: the imaginary kinetic term, absent in the single-generator framework of Papers XV–XVII, emerges from the pairing of forward and backward processes. The remaining open problem is no longer the kinetic sector but the nonlinear state-dependent potential away from equilibrium. The resulting nonlinear correction is not interpreted here as a modification of quantum mechanics, but as a structurally derived residual whose physical status is deferred to subsequent work.

1 Introduction

Paper XVII [Smart, 2026d] proved the dissipative obstruction: any generator of the form $L_{\mathrm{BK}} + iM$, where L_{BK} is the backward Kolmogorov generator from the closure Langevin process, is dissipative in the equilibrium-weighted inner product regardless of the imaginary potential iM. This ruled out the single-generator approach of Papers XV–XVI [Smart, 2026b,c] as a route to unitary evolution.

Paper XVII classified three candidate routes past the obstruction and identified Route B — the Nelson-type forward/backward pairing — as the most promising, because it uses the existing stochastic closure process and its time-reversal without introducing new canonical variables.

The present paper carries out this construction.

The problem

The goal is not to assume Schrödinger structure, but to determine whether it *emerges* from the paired forward/backward closure processes. Specifically:

(P10) **The current-velocity problem.** Can the closure framework supply a non-stationary dynamical equation for the current velocity v?

(P11) **The Schrödinger-recovery problem.** Does the Nelson wavefunction $\Psi_N = \sqrt{\rho}\, e^{iS/\sigma^2}$ satisfy a Schrödinger-type equation?

Scope and epistemic status

This paper derives the closure-Nelson wave equation and identifies the regime in which it reduces to the Schrödinger equation. Where results are derived, they are stated as propositions; where they depend on approximations (linearisation near equilibrium), that status is explicit. More precisely, the paper derives an exact nonlinear closure-Nelson wave equation and identifies the regime in which its leading-order dynamics reduces to the linear Schrödinger equation.

2 Forward and Backward Closure Processes

2.1 The forward process

The closure Langevin process of Paper XIV [Smart, 2026a] is:

$$d\psi_t = b_+(\psi_t)\, dt + \sigma\, dW_t^+, \qquad b_+ = -\nabla\mathcal{F} \tag{1}$$

where $\mathcal{F} \in C^2(\mathcal{S}_{\mathrm{adm}})$ is the closure functional and $\sigma > 0$ is the noise amplitude. The probability density $\rho(\psi, t)$ satisfies the forward Fokker–Planck equation:

$$\partial_t \rho = \nabla \cdot (\rho\, \nabla\mathcal{F}) + \frac{\sigma^2}{2}\Delta\rho \tag{2}$$

The stationary distribution is $\rho_{\mathrm{st}} = Z^{-1} \exp(-2\mathcal{F}/\sigma^2)$.

2.2 The backward process

The time-reversed diffusion has backward drift:

$$b_-(\psi, t) = b_+(\psi) - \sigma^2 \nabla \log \rho(\psi, t) = -\nabla \mathcal{F} - \sigma^2 \nabla \log \rho \tag{3}$$

2.3 Nelson velocities

Define:

$$v = \frac{b_+ + b_-}{2} = -\nabla \mathcal{F} - \frac{\sigma^2}{2} \nabla \log \rho \qquad \text{(current velocity)} \tag{4}$$

$$u = \frac{b_+ - b_-}{2} = \frac{\sigma^2}{2} \nabla \log \rho \qquad \text{(osmotic velocity)} \tag{5}$$

The current velocity represents the net probabilistic flow; the osmotic velocity represents the diffusive spreading.

Remark 1 (Stationarity check)**.** *At the Paper XIV equilibrium $\rho = \rho_{\text{st}}$: $\nabla \log \rho_{\text{st}} = -2\nabla \mathcal{F}/\sigma^2$, giving $v_{\text{st}} = 0$ and $u_{\text{st}} = -\nabla \mathcal{F}$. The vanishing current velocity at equilibrium is consistent with the ground-state condition in Nelson mechanics.*

3 Continuity Equation

Proposition 2 (Continuity equation)**.** *The probability density ρ and current velocity v satisfy:*

$$\partial_t \rho + \nabla \cdot (\rho\, v) = 0 \tag{6}$$

Proof. From (4), $\rho\, v = -\rho\, \nabla \mathcal{F} - (\sigma^2/2)\nabla \rho$, so:

$$\nabla \cdot (\rho\, v) = -\nabla \cdot (\rho\, \nabla \mathcal{F}) - \frac{\sigma^2}{2} \Delta \rho \tag{7}$$

Comparing with the Fokker–Planck equation (2): $\partial_t \rho = \nabla \cdot (\rho\, \nabla \mathcal{F}) + (\sigma^2/2)\Delta \rho = -\nabla \cdot (\rho\, v)$. □

This is the first Madelung equation. It is exact and requires no approximation. It is also the first place where the paired-process construction departs structurally from the single-generator framework of Paper XVI: the current velocity v involves both forward and backward drifts, not just the forward drift b_+.

4 Phase Function and Irrotationality

Proposition 3 (Irrotational current velocity)**.** *The current velocity v defined in (4) is a gradient field:*

$$v = \nabla S, \qquad S(\psi, t) = -\mathcal{F}(\psi) - \frac{\sigma^2}{2} \log \rho(\psi, t) \tag{8}$$

Proof. Both $-\nabla \mathcal{F}$ and $-(\sigma^2/2)\nabla \log \rho$ are gradients, so their sum is a gradient. The phase function S is determined (up to a spatially constant, time-dependent function) by $\nabla S = v$. □

Remark 4. *Irrotationality is not an assumption — it is a consequence of the closure framework, in which both the drift $b_+ = -\nabla\mathcal{F}$ and the density correction $\nabla \log \rho$ are gradient fields. Because irrotationality is inherited from the gradient structure of both the closure drift and the density correction, no separate phase ansatz is needed: the phase function S is forced by the closure geometry. This is a structural advantage of the closure framework over generic stochastic processes, where v need not be irrotational.*

Writing $R = \sqrt{\rho}$, the phase can be expressed as:

$$S = -\mathcal{F} - \sigma^2 \log R \tag{9}$$

5 The Closure Madelung Equations

We now derive the second Madelung equation from the Fokker–Planck dynamics.

Proposition 5 (Closure Madelung pair)**.** *The pair (ρ, S) satisfies:*

$$\partial_t \rho + \nabla \cdot (\rho \nabla S) = 0 \tag{10}$$

$$\partial_t S + \frac{|\nabla S|^2}{2} - \sigma^2 V_{\mathrm{W}} + \frac{\sigma^4}{2}\frac{\Delta\sqrt{\rho}}{\sqrt{\rho}} = 0 \tag{11}$$

where $V_{\mathrm{W}} = |\nabla\mathcal{F}|^2/(2\sigma^2) - \Delta\mathcal{F}/2$ is the Witten potential of Paper XVI.

Proof. Equation (10) is Proposition 2 with $v = \nabla S$.

For (11): from $S = -\mathcal{F} - \sigma^2 \log R$ with $R = \sqrt{\rho}$, the time derivative is $\partial_t S = -\sigma^2 \partial_t \log R = -(\sigma^2/2)\partial_t \log \rho$. From the Fokker–Planck:

$$\frac{\partial_t \rho}{\rho} = \Delta\mathcal{F} + \nabla\mathcal{F} \cdot \nabla \log \rho + \frac{\sigma^2}{2}(\Delta \log \rho + |\nabla \log \rho|^2) \tag{12}$$

Using $\nabla \log \rho = 2\nabla \log R$ and $\Delta \log \rho = 2\Delta \log R$:

$$\partial_t S = -\frac{\sigma^2}{2}\Delta\mathcal{F} - \sigma^2 \nabla\mathcal{F} \cdot \nabla \log R - \frac{\sigma^4}{2}\Delta \log R - \sigma^4 |\nabla \log R|^2 \tag{13}$$

The kinetic term is $|v|^2/2 = |\nabla\mathcal{F}|^2/2 + \sigma^2 \nabla\mathcal{F} \cdot \nabla \log R + (\sigma^4/2)|\nabla \log R|^2$.

Adding, and using $\Delta \log R + |\nabla \log R|^2 = \Delta R/R$:

$$\begin{aligned} \partial_t S + \frac{|v|^2}{2} &= \frac{|\nabla\mathcal{F}|^2}{2} - \frac{\sigma^2}{2}\Delta\mathcal{F} - \frac{\sigma^4}{2}\frac{\Delta R}{R} \\ &= \sigma^2 V_{\mathrm{W}} - \frac{\sigma^4}{2}\frac{\Delta\sqrt{\rho}}{\sqrt{\rho}} \end{aligned} \tag{14}$$

Rearranging gives (11). □

5.1 Comparison with the Schrödinger Madelung equations

The standard Schrödinger equation $i\hbar\partial_t\psi = -(\hbar^2/2m)\Delta\psi + V\psi$ is equivalent to the Madelung pair:

$$\partial_t \rho + \nabla \cdot (\rho \nabla S/m) = 0 \tag{15}$$

$$\partial_t S + \frac{|\nabla S|^2}{2m} + V - \frac{\hbar^2}{2m}\frac{\Delta\sqrt{\rho}}{\sqrt{\rho}} = 0 \tag{16}$$

The closure Madelung equation (11) has the same structure as (16) with:

- effective mass $m_{\text{eff}} = 1$ (from $|v|^2/2$ rather than $|v|^2/(2m)$),
- effective potential $V_{\text{eff}} = -\sigma^2 V_{\text{W}}$ (with opposite sign to the usual convention),
- quantum potential $Q = +(\sigma^4/2)\Delta\sqrt{\rho}/\sqrt{\rho}$ (with **opposite sign** to the Schrödinger convention).

The sign reversal of the quantum potential is the key structural difference between the closure Madelung system and the standard Schrödinger Madelung system. However, this should not be read as a failure of the pairing construction. Rather, it identifies the precise way in which the closure-paired dynamics differs from exact linear quantum mechanics away from equilibrium: the sign discrepancy is absorbed into a nonlinear state-dependent potential in the recombined wave equation (Section 6), and vanishes entirely in the near-equilibrium linearisation (Section 7).

6 The Closure-Nelson Wave Equation

Recombining ρ and S into the Nelson wavefunction:

$$\Psi_N = \sqrt{\rho}\, e^{iS/\sigma^2} \tag{17}$$

Proposition 6 (Closure-Nelson wave equation)**.** *The Madelung pair* (10)–(11) *is equivalent to the nonlinear wave equation:*

$$\boxed{i\sigma^2\, \partial_t \Psi_N = -\frac{\sigma^4}{2}\Delta\Psi_N + U[\Psi_N]\,\Psi_N} \tag{18}$$

where the state-dependent potential is:

$$U[\Psi_N] = -\sigma^2 V_{\text{W}} + \sigma^4 \frac{\Delta|\Psi_N|}{|\Psi_N|} \tag{19}$$

Proof. The standard Madelung correspondence states that $i\sigma^2\partial_t\Psi = -(\sigma^4/2)\Delta\Psi + U\Psi$ with $\Psi = Re^{iS/\sigma^2}$ is equivalent to:

$$\partial_t\rho + \nabla\cdot(\rho\,\nabla S) = 0 \tag{20}$$

$$\partial_t S + \frac{|\nabla S|^2}{2} + U - \frac{\sigma^4}{2}\frac{\Delta R}{R} = 0 \tag{21}$$

Comparing the second equation with (11):

$$U - \frac{\sigma^4}{2}\frac{\Delta R}{R} = -\sigma^2 V_{\text{W}} + \frac{\sigma^4}{2}\frac{\Delta R}{R} \tag{22}$$

Solving: $U = -\sigma^2 V_{\text{W}} + \sigma^4\Delta R/R = -\sigma^2 V_{\text{W}} + \sigma^4\Delta|\Psi_N|/|\Psi_N|$. □

Remark 7 (Nature of the equation)**.** *Equation* (18) *is a nonlinear Schrödinger equation: the potential U depends on the state Ψ_N through $\Delta|\Psi_N|/|\Psi_N|$. It has the correct kinetic structure ($-(\sigma^4/2)\Delta$ with the imaginary prefactor $i\sigma^2$) but a state-dependent potential. The nonlinearity enters through the quantum-potential sign reversal identified in Section 5.*

The paired-process construction has therefore succeeded in generating the missing imaginary kinetic structure that was impossible in the single-generator setting of Papers XVI–XVII. What remains is not the kinetic problem but the residual state dependence of the effective potential.

Remark 8 (Why this evades the XVII obstruction)**.** *Paper XVII ruled out unitarity for generators of the single-process Kolmogorov form $L_{\mathrm{BK}} + iM$. The present construction does not modify that form; it leaves it behind. The forward/backward pairing produces a different composite dynamics: the current velocity $v = (b_+ + b_-)/2$ involves both forward and backward drifts, and the resulting kinetic structure is no longer governed by a single dissipative Kolmogorov generator. The imaginary kinetic term $-(\sigma^4/2)\Delta$ therefore emerges from the pairing structure itself, not from any modification of a single Kolmogorov generator, and is consistent with the obstruction theorem of Paper XVII.*

7 Near-Equilibrium Linearisation

The nonlinear potential $U[\Psi_N]$ simplifies dramatically near the Paper XIV equilibrium.

Proposition 9 (Equilibrium and near-equilibrium Schrödinger regime)**.** *The closure-Nelson wave equation (18) admits a Schrödinger regime in two stages:*

1. ***Exact at equilibrium.*** *At $\rho = \rho_{\mathrm{st}}$, the nonlinear potential evaluates to the constant:*
$$U[\Psi_{\mathrm{st}}] = \sigma^2 V_{\mathrm{W}} \tag{23}$$
and the wave equation reduces exactly to $i\sigma^2\partial_t\Psi = \sigma^2 H_{\mathrm{W}}\Psi$.

2. ***Leading-order near equilibrium.*** *For perturbations $\Psi = \Psi_{\mathrm{st}}(1+\varepsilon\eta)$ with $\varepsilon \ll 1$, the nonlinear correction is $O(\varepsilon)$, and the wave equation linearises, to leading order in ε, to:*
$$\boxed{i\,\partial_t\Psi = H_{\mathrm{W}}\,\Psi} \tag{24}$$
where $H_{\mathrm{W}} = -(\sigma^2/2)\Delta + V_{\mathrm{W}}$ is the non-negative Witten Hamiltonian of Paper XVI.

Proof. At stationarity, $R_{\mathrm{st}} = \sqrt{\rho_{\mathrm{st}}} \propto e^{-\mathcal{F}/\sigma^2}$, so:
$$\frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} = \frac{|\nabla\mathcal{F}|^2}{\sigma^4} - \frac{\Delta\mathcal{F}}{\sigma^2} = \frac{2V_{\mathrm{W}}}{\sigma^2} \tag{25}$$
Substituting into (19): $U[\Psi_{\mathrm{st}}] = -\sigma^2 V_{\mathrm{W}} + \sigma^4 \cdot 2V_{\mathrm{W}}/\sigma^2 = \sigma^2 V_{\mathrm{W}}$.

The full wave equation (18) at stationarity is:
$$i\sigma^2\partial_t\Psi = -\frac{\sigma^4}{2}\Delta\Psi + \sigma^2 V_{\mathrm{W}}\,\Psi = \sigma^2\left(-\frac{\sigma^2}{2}\Delta + V_{\mathrm{W}}\right)\Psi = \sigma^2 H_{\mathrm{W}}\Psi \tag{26}$$
Dividing by σ^2 gives $i\partial_t\Psi = H_{\mathrm{W}}\Psi$.

For the linearisation: write $\Psi = \Psi_{\mathrm{st}}(1+\varepsilon\eta)$ for small perturbation η. Then $|\Psi| = R_{\mathrm{st}}(1+\varepsilon\,\mathrm{Re}(\eta)) + O(\varepsilon^2)$, and:
$$\frac{\Delta|\Psi|}{|\Psi|} = \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} + O(\varepsilon) = \frac{2V_{\mathrm{W}}}{\sigma^2} + O(\varepsilon) \tag{27}$$
So $U = \sigma^2 V_{\mathrm{W}} + O(\varepsilon)$, and the linearised equation is $i\partial_t\Psi = H_{\mathrm{W}}\Psi$ to leading order. □

Thus the exact equilibrium state furnishes not only the ground state of the closure dynamics, but also the base point around which linear quantum evolution emerges.

Remark 10 (Interpretation)**.** *The Witten Hamiltonian H_{W} appeared in Paper XVI as the real part of the drift-free generator, where it produced dissipation. Here, via the Nelson pairing, it reappears as the quantum Hamiltonian governing oscillatory dynamics. The same operator that drives dissipation in the single-process framework drives oscillation in the paired-process framework.*

Remark 11 (On the non-triviality of the linear limit)**.** *It is true that any nonlinear dynamical system admits a linearisation at a fixed point. The non-trivial result here is not the existence of a linearisation, but the specific structure of that linearisation: the linearised generator is exactly the Witten Hamiltonian H_{W}, which is derived from the closure functional $\mathcal{F}$ and the noise scale σ rather than chosen or postulated. The content is therefore not merely that a linear regime exists, but that it reproduces a specific quantum Hamiltonian determined by the closure geometry.*

Remark 12 (Effective parameters)**.** *Within this normalisation, the linearised equation $i\partial_t\Psi = H_{\mathrm{W}}\Psi$ may be written in Schrödinger form with effective parameters:*

- *$\hbar_{\mathrm{eff}} = 1$ in these units (or σ^2 before the division),*
- *effective mass $m_{\mathrm{eff}} = 1/\sigma^2$ (from $-(\sigma^2/2)\Delta = -\hbar_{\mathrm{eff}}^2/(2m_{\mathrm{eff}})\Delta$),*
- *effective potential $V_{\mathrm{W}} = |\nabla\mathcal{F}|^2/(2\sigma^2) - \Delta\mathcal{F}/2$.*

These are read off from the closure functional $\mathcal{F}$ and the noise scale σ; no additional postulates are needed within the linearised regime.

8 Worked Example: Quadratic Closure Functional

Example 13 (Quadratic closure)**.** *Let $\mathcal{F}(\psi) = \frac{1}{2}k\psi^2$ in one dimension.*

Nelson velocities

$b_+ = -k\psi$, $\rho_{\mathrm{st}} = Z^{-1}e^{-k\psi^2/\sigma^2}$, $\nabla\log\rho_{\mathrm{st}} = -2k\psi/\sigma^2$.

At stationarity: $v = 0$, $u = -k\psi$. The phase $S = -k\psi^2/2 - \sigma^2\log R_{\mathrm{st}} = 0$ (constant, since $R_{\mathrm{st}} = Z^{-1/2}e^{-k\psi^2/(2\sigma^2)}$ and $-k\psi^2/2 - \sigma^2(-k\psi^2/(2\sigma^2)) = 0$).

Witten potential and Hamiltonian

$V_{\mathrm{W}} = k^2\psi^2/(2\sigma^2) - k/2$. *The Witten Hamiltonian is:*

$$H_{\mathrm{W}} = -\frac{\sigma^2}{2}\partial_\psi^2 + \frac{k^2\psi^2}{2\sigma^2} - \frac{k}{2} \tag{28}$$

This is a shifted quantum harmonic oscillator with spectrum $E_n = nk$, $n = 0, 1, 2, \ldots$

Linearised Schrödinger dynamics

The linearised equation $i\partial_t\Psi = H_{\mathrm{W}}\Psi$ has eigenstates ϕ_n satisfying $H_{\mathrm{W}}\phi_n = nk\,\phi_n$. The ground state $\phi_0 \propto e^{-k\psi^2/(2\sigma^2)}$ is stationary. Excited states oscillate:

$$\Psi_n(t) = e^{-inkt}\phi_n \tag{29}$$

with fundamental frequency $\omega = k$. The energy gap is $\Delta E = k$ — determined entirely by the closure functional.

Comparison with Paper XVII

In Paper XVII's dissipative framework, the same excited states ϕ_n dissipate to the ground state at rate $2nk$. In the Nelson-paired framework, they oscillate with frequency nk. The Witten Hamiltonian governs both: dissipation rate $= 2E_n$, oscillation frequency $= E_n$.

Table 1: Quadratic example: dissipative vs. oscillatory dynamics.

Feature	Paper XVII (single process)	Paper XVIII (paired processes)
Generator	$-H_{\mathrm{W}} + iM$	iH_{W} (linearised)
Ground state	$e^{-k\psi^2/(2\sigma^2)}$, stable	$e^{-k\psi^2/(2\sigma^2)}$, stable
Excited states	Dissipate at rate $2nk$	Oscillate at frequency nk
Norm	Decreasing	Preserved (linearised)

The quadratic example shows the programme transition in its cleanest form: the same Witten operator that appears as a dissipation generator in the single-process framework (Paper XVII) becomes the oscillation generator in the paired-process linearised regime.

9 Conditions for Exact Schrödinger Recovery

The closure-Nelson wave equation (18) is exactly Schrödinger (i.e. the potential U is state-independent) if and only if:

$$\sigma^4 \frac{\Delta|\Psi_N|}{|\Psi_N|} = \text{function of } \psi \text{ only, independent of } t \tag{30}$$

This is a strong condition. In the present framework, the physically relevant instance occurs in the near-equilibrium regime: for small perturbations $\Psi = \Psi_{\mathrm{st}}(1 + \varepsilon\eta)$, the nonlinear term is $O(\varepsilon)$ and the linearised dynamics is exactly Schrödinger (Proposition 9). For generic non-equilibrium states, the nonlinear correction is present. The correction can be characterised as:

$$\delta U = \sigma^4 \left(\frac{\Delta|\Psi_N|}{|\Psi_N|} - \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} \right) = \sigma^4 \left(\frac{\Delta R}{R} - \frac{2V_{\mathrm{W}}}{\sigma^2} \right) \tag{31}$$

This vanishes when R has the same curvature structure as R_{st} — i.e. when the density profile is close to the equilibrium Gaussian (in the quadratic case) or more generally when ρ is close to ρ_{st}. Accordingly, the generic paired closure dynamics is nonlinear, while exact linear Schrödinger evolution appears as a distinguished regime selected by equilibrium structure or special modulus profiles.

10 Interpretation of the Nonlinear Residual

The closure-Nelson wave equation (18) differs from the linear Schrödinger equation by the state-dependent term:

$$\delta U[\Psi_N] = \sigma^4 \frac{\Delta|\Psi_N|}{|\Psi_N|} \tag{32}$$

This term vanishes at equilibrium and is small for perturbations around ρ_{st}, but is generically nonzero for non-equilibrium states.

Structural origin

The residual term arises from the same quantity that produces the quantum potential in the Madelung formulation:

$$Q = \frac{\hbar^2}{2m} \frac{\Delta\sqrt{\rho}}{\sqrt{\rho}} \tag{33}$$

In standard quantum mechanics, Q appears with a sign that cancels nonlinearity when recombined into the Schrödinger equation. In the closure framework, the sign inherited from the dissipative forward process leads to a non-cancelling contribution, producing the residual δU.

What the residual measures

The residual encodes the deviation of the density profile from the equilibrium structure:

$$\delta U = \sigma^4 \left(\frac{\Delta R}{R} - \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} \right) \tag{34}$$

Thus:

- $\delta U = 0$ for equilibrium density profiles,
- $\delta U \neq 0$ for non-equilibrium configurations,
- the magnitude of δU encodes the curvature deviation of ρ from ρ_{st}.

Status

At this stage, the residual term is identified as a structural consequence of the closure pairing, not interpreted as a modification of quantum mechanics. Whether it represents:

- a physical nonlinear correction to linear quantum mechanics,
- a gauge artefact removable by transformation,
- or a structurally meaningful quantity with independent physical content,

remains an open question and is deferred to subsequent work. In this sense, the residual is best understood at present as the closure-sector remainder after linear quantum behaviour has been recovered, rather than as a defect of the paired construction.

11 Reconciliation with Papers XV–XVII

11.1 Paper XV: phase and interference

Paper XV introduced the complexified path integral with amplitude $\mathcal{A} = e^{-S/\sigma^2 + i\Theta/\sigma^2}$. The Nelson wavefunction $\Psi_N = \sqrt{\rho}\, e^{iS/\sigma^2}$ is a different object: it is constructed from the density and phase of the paired processes, not from the path-integral amplitude. However, at equilibrium both reduce to $\Psi \propto e^{-\mathcal{F}/\sigma^2}$ (the same ground state), and both produce the squared-modulus probability structure $P = |\Psi|^2 = \rho$.

11.2 Paper XVI: evolution equation

The closure evolution equation of Paper XVI, $\partial_t \Psi = (\sigma^2/2)\Delta\Psi - \nabla\mathcal{F} \cdot \nabla\Psi + (i/\sigma^2)\mathcal{F}\Psi$, governs the single forward process. The Nelson wave equation (18) governs the paired forward-backward system. They share the same equilibrium state but have different generator structures: the former is dissipative (Paper XVII), the latter is oscillatory (in the linearised regime).

11.3 Paper XVII: dissipative obstruction

The obstruction theorem applies to generators of the single-process form $L_{\mathrm{BK}} + iM$. The Nelson construction evades it by not using a single Kolmogorov generator: the current velocity $v = (b_+ + b_-)/2$ involves both forward and backward drifts, and the resulting dynamics is not of the $L_{\mathrm{BK}} + iM$ form. The imaginary kinetic term emerges from the pairing, not from a modification of the single-process generator. In this sense, XVIII should be read not as a contradiction of XVII, but as the constructive completion of the route XVII left open.

12 Discussion and Limitations

12.1 What is derived

Exact results:

- The continuity equation for the current velocity v (Proposition 2).
- Irrotationality of v and the existence of the phase function S (Proposition 3), a structural consequence of the closure gradient drift.
- The closure Madelung pair (Proposition 5).
- The closure-Nelson wave equation (18) (Proposition 6): exact, but nonlinear.

Exact but regime-specific:

- At exact equilibrium, $U[\Psi_{\mathrm{st}}] = \sigma^2 V_{\mathrm{W}}$ and the wave equation is exactly the Schrödinger equation with the Witten Hamiltonian (Proposition 9, stage 1).

Leading-order / approximate:

- Near-equilibrium linearisation to $i\partial_t \Psi = H_{\mathrm{W}}\Psi$, valid to leading order in perturbations around ρ_{st} (Proposition 9, stage 2).
- The effective parameters $m_{\mathrm{eff}} = 1/\sigma^2$ and $V_{\mathrm{eff}} = V_{\mathrm{W}}$ are read off within the linearised regime (Remark 12).

12.2 What is conditional

- Exact Schrödinger form holds at equilibrium, and the near-equilibrium linearisation is valid to leading order for sufficiently small perturbations. Away from equilibrium, the exact closure-Nelson dynamics remains nonlinear through the correction $\sigma^4 \Delta|\Psi|/|\Psi|$.

- The linearisation requires the perturbation $\varepsilon\eta$ to be small relative to Ψ_{st}. The range of validity is not quantified.
- The identification $\hbar_{\mathrm{eff}} = \sigma^2$ (before unit absorption) is a structural correspondence within this construction, not a derived physical identification.

12.3 What is not achieved

- **No exact global Schrödinger recovery.** The full closure-Nelson equation is nonlinear; the Schrödinger equation is linear.
- **No Hilbert-space axioms.** The construction produces a wave equation, not a full quantum-mechanical framework with measurement postulates.
- **No physical derivation of the backward process.** The time-reversed process is defined mathematically; whether it has independent physical content within the closure programme is not addressed.
- **The quantum-potential sign reversal.** The closure Madelung equation has the quantum potential with the opposite sign to standard quantum mechanics. This is absorbed into the nonlinear potential U but represents a genuine structural difference.
- **No exact identification with standard quantum mechanics away from equilibrium.** The paired closure dynamics agrees with linear Schrödinger evolution only in the equilibrium and near-equilibrium regime.

12.4 Open questions

- Can the nonlinear correction be absorbed by a redefinition or gauge transformation?
- Does the nonlinear Schrödinger equation (18) have additional conserved quantities?
- Can the range of validity of the linearisation be characterised (e.g. in terms of Rényi divergence from equilibrium)?
- Is there a variational principle for the closure Madelung pair that would select the dynamics independently of the Nelson construction?

13 Conclusion

Paper XVIII changes the location of the remaining problem in the VFD quantum-recovery programme.

Exact results. Starting from the forward closure Langevin process (Paper XIV) and its time-reversal, the Nelson pairing produces:

1. an exact continuity equation for the current velocity,
2. an automatically irrotational phase function (forced by the closure gradient structure),
3. an exact closure Madelung pair with a quantum-potential-like term,

4. a nonlinear Schrödinger-type wave equation (18) with the correct imaginary kinetic structure.

Equilibrium Schrödinger recovery. At the Paper XIV equilibrium, the nonlinear potential evaluates exactly to $\sigma^2 V_{\mathrm{W}}$, and the wave equation becomes exactly the Schrödinger equation with the Witten Hamiltonian:

$$i\,\partial_t \Psi = H_{\mathrm{W}} \Psi = \left(-\frac{\sigma^2}{2} \Delta + V_{\mathrm{W}} \right) \Psi \tag{35}$$

The Witten Hamiltonian — which appeared in Paper XVI as the dissipative real part of the drift-free generator — now reappears as the quantum Hamiltonian governing oscillatory dynamics. For perturbations around equilibrium, this Schrödinger equation persists to leading order.

What remains. The closure-Nelson construction resolves the kinetic-sector obstruction identified in Paper XVII: the imaginary kinetic term emerges from the forward/backward pairing. The remaining discrepancy is confined to the potential sector: a residual state-dependent term $\sigma^4 \Delta |\Psi| / |\Psi|$ (Section 10) that vanishes at equilibrium and is small for near-equilibrium perturbations.

This shifts the problem from the existence of quantum dynamics to the interpretation of this residual term. The closure framework therefore reproduces standard linear quantum mechanics in a well-defined equilibrium-centred regime while yielding a controlled, structurally derived deviation away from that regime. Determining the nature of this deviation — whether it is a physical nonlinear correction, a gauge artefact, or a structurally meaningful quantity — is the next stage of the programme.

Programme arc.

- Papers XII–XIV: dissipative closure dynamics,
- Paper XV: phase and interference,
- Paper XVI: evolution equation and structural gap,
- Paper XVII: dissipative obstruction theorem,
- Paper XVIII: Nelson pairing, near-equilibrium Schrödinger recovery, and identification of the nonlinear residual as the next open problem.

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